

## CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 10 · LECTURE NOTES

# Burst Photography

*What your phone actually does.*

## MODULE OUTCOMES

- Explain why bursts beat single exposures.
- Implement robust alignment for a burst.
- Implement merging that rejects misaligned content.
- Explain the complete modern phone pipeline.
- Explain super-resolution from a handheld burst.

## 1 Why bursts

### MORE FRAMES IS MORE PHOTONS WITHOUT MORE BLUR

**A long exposure collects more photons**, which is what reduces shot noise (Module 01) — **and it blurs**, from hand tremor and from anything in the scene that moves.

**Many short exposures collect the same total photons with no motion blur in any individual frame.** Each frame is noisy, and none is blurred.

**Averaging  $n$  aligned frames improves signal-to-noise by  $\sqrt{n}$**  — the signal adds linearly and the independent noise adds in quadrature. Sixteen frames gives a factor of four, which is two stops.

**So sixteen short exposures are equivalent to one sixteen-times longer exposure, and sharp.** The cost is the alignment, which is computation — and computation is cheap and improving quickly, while sensor area in a phone is fixed by the thickness of the device.

| Constraint                         | Consequence                                                                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Phone sensors are physically small | Few photons per pixel, so images are <b>shot-noise limited</b> in all but bright light (Module 01).                                              |
| Apertures are fixed and small      | No optical route to gathering more light — the usual photographic lever is unavailable.                                                          |
| Phones are handheld                | <b>Long exposures blur.</b> The usual alternative lever is also unavailable.                                                                     |
| Computation is abundant            | <b>And it improves faster than optics or sensor physics.</b> A phone has a capable image signal processor and several seconds of thermal budget. |
| Users press one button             | Whatever happens must be fully automatic, with no bracketing choices or post-processing expected.                                                |

**Every constraint points the same way:** capture many frames and compute the photograph. This is why every manufacturer converged on essentially the same approach within a few years — it was not a design preference but the only direction available.

## 2 Alignment

### BURST ALIGNMENT IS EASIER THAN PANORAMA ALIGNMENT

**The frames are nearly identical.** Same exposure, same scene, captured milliseconds apart under the same light — which is precisely the condition Module 03 noted that bracketed exposures do not satisfy.

**The displacements are small** — hand tremor over a few tens of milliseconds moves the frame by a few pixels, not by a large rotation.

**So the full feature-detection and RANSAC machinery of Module 07 is unnecessary.** Hierarchical block matching — directly minimising the difference between tiles — is sufficient, far simpler, and much faster.

**Align tiles rather than whole frames.** A single global transform captures camera motion but not the fact that a person in the scene moved independently. **A per-tile displacement field handles both**, and it is what makes robust merging possible in §2.1.

```
Build a Gaussian pyramid of the reference and each alternate frame.
For each level, coarse to fine:
  for each tile:
    search a small window around the previous level's estimate
    minimise L1 or L2 distance to the reference tile
    upsample the displacement to the next level
```

Result: a PER-TILE displacement field, not a global transform.

**Tiles** because the camera rotates (a global effect) *and* things in the scene move (a local one), and only a per-tile field captures both. **Coarse-to-fine** (Module 02) because it keeps the search window small at every level, so a displacement of fifty pixels costs barely more to find than one of five.

**MERGING MUST REJECT WHAT DOES NOT MATCH**

**Alignment will fail somewhere.** A region occluded in one frame and not another, a subject that moved, a patch of sky with no texture to match against — these are normal, not exceptional.

**Averaging a misaligned tile produces ghosting**, which is a worse artefact than the noise the averaging was intended to remove. Naive averaging is therefore not an option.

**So merge with a robustness test:** compare each aligned frame against the reference and down-weight contributions wherever they disagree by more than the noise model (Module 01) can explain. Where frames agree, average fully; where they do not, fall back toward the reference alone.

**HDR+ performs this in the frequency domain**, applying a Wiener-like shrinkage per frequency band rather than a binary per-pixel decision. **This handles partial misalignment gracefully** — a tile may be well aligned at low frequencies and poorly at high ones, and a frequency-wise decision uses what is usable rather than discarding the tile.

### 3 The pipeline, end to end

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1. CAPTURE 10-15 RAW frames, all UNDEREXPOSED  
-> highlights never clip; all frames short, so none blur
2. SELECT a reference frame (sharpest, usually near the middle)
3. ALIGN all others to it, per tile, hierarchically
4. MERGE with robustness -- reject tiles that disagree
5. Then the classical pipeline on a far better input:  
black level, white balance, DEMOSAIC (Module 05)  
chroma denoise, local tone map (Module 04)  
sharpen, colour correct, ENCODE

#### MERGE BEFORE DEMOSAICING, IN LINEAR SPACE

**Merging must happen on linear data** (Module 01). Averaging gamma-encoded values does not give the encoding of the average radiance, and the error is largest exactly where the signal is weakest.

**And it must happen before demosaicing.** Demosaicing is an interpolation — it invents two-thirds of the colour data at every pixel (Module 05). **Averaging sixteen frames' worth of invented data is far less useful than averaging sixteen frames' worth of measurements**, and the invented values are correlated with the measurement errors that produced them.

**Merging raw also means the merged result carries more bits than any input frame.** Averaging sixteen 10-bit frames yields roughly 12 bits of real precision, which the subsequent tone mapping can use — and which is part of why night mode images tolerate aggressive shadow lifting.

**So the burst machinery sits between the sensor and the classical pipeline**, not after it. Module 05's nine steps are unchanged; they simply operate on a much better input.

### 4 Super-resolution from hand tremor

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**HAND SHAKE SAMPLES BETWEEN THE PIXELS**

**Natural hand tremor displaces the sensor by sub-pixel amounts between frames** — fractions of a pixel, essentially at random.

**So each frame samples the scene at slightly different positions.** Frame one measures the scene at one grid of points, frame two at a grid offset by 0.3 pixels, and so on.

**Combining them onto a finer grid recovers detail beyond any single frame's sampling limit.** And because the Bayer pattern moves with the sensor, different colour filters land on different scene points across the burst — so **every colour is directly measured somewhere, and demosaicing becomes unnecessary.** The interpolation of Module 05 is replaced by measurement.

**Hand tremor, normally a defect to be corrected, is the mechanism that makes it work.** Google's super-resolution zoom *deliberately introduces* sensor motion via the optical stabiliser when the phone is on a tripod, because without tremor every frame samples identically and there is nothing to recover. **This is a genuinely delightful result** and a good example of a constraint turning out to be a resource.

- **It cannot recover clipped highlights.** Nothing can (Module 03) — which is exactly why the frames are deliberately underexposed.
- **It cannot handle fast subject motion.** A moving person is taken from the reference frame alone, with that single frame's noise — so a night photograph of a moving subject is noisy precisely where the subject is, which is a recognisable signature.
- **It struggles in near-darkness,** where even the sum of fifteen frames is photon-starved and the result is dominated by whatever the denoiser decided to do.
- **And the artefacts are visible once you know them:** oversharpened edges with faint halos, noise that is unnaturally even across the frame, and detail that is **slightly too clean** — foliage and hair in particular take on a characteristic smoothed-and-resharpened quality.
- **Which is the lead-in to Modules 12 and 13,** where the question of what was measured and what was synthesised becomes the subject rather than a footnote.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                                                              | What it covers for this module                                                                                                                             |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hasinoff et al. — Burst photography for high dynamic range and low-light imaging (free, with dataset) | <b>The HDR+ paper and the raw burst dataset.</b> This module is essentially a reading of it, and the dataset lets you implement the pipeline on real data. |
| Wronski et al. — Handheld Multi-Frame Super-Resolution (free)                                         | The §4 result, including the deliberate motion on a tripod.                                                                                                |

| Resource                                                           | What it covers for this module                                                     |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Levoy & Pritch — Night Sight blog posts (free)                     | How the pipeline was extended to very low light, written by the people who did it. |
| Liba et al. — Handheld Mobile Photography in Very Low Light (free) | Motion metering, exposure selection, and the limits in §4.                         |

## Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Capture a burst of 16 raw frames handheld in dim light. Align and average them and compare against a single frame and against a long exposure.
- 2 Measure the SNR improvement against frame count and confirm the  $\sqrt{n}$  relationship on real data.
- 3 Implement hierarchical tile alignment on the HDR+ dataset. Visualise the displacement field.
- 4 Average without any robustness test on a burst containing a moving subject, and photograph the resulting ghosting.
- 5 Implement a robustness-weighted merge and show the ghosting removed.
- 6 Merge before and after demosaicing on the same burst and compare the result.
- 7 Merge in linear space and in gamma space and quantify the difference.
- 8 Implement a simple multi-frame super-resolution: align a burst to sub-pixel accuracy, accumulate onto a  $2\times$  grid, and compare against upsampling a single frame.
- 9 Find the burst-processing artefacts of §4 in photographs from your own phone. Foliage and hair are the places to look.

## Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why do many short exposures beat one long one, and by how much?
- 2 Give five constraints that made burst photography inevitable for phones.
- 3 Why is burst alignment easier than panorama alignment, and what method suffices?
- 4 Why align per tile rather than globally?
- 5 Why must merging reject disagreement, and what does HDR+ do in the frequency domain?
- 6 Give the eight steps of the modern phone pipeline.
- 7 Why must merging precede demosaicing and happen in linear space?

- 8 How does hand tremor enable super-resolution, and what happens on a tripod?
- 9 Give four things burst processing cannot do, and three visible artefacts.